

CHE 201 Spring 2020 Quiz 1

Roxanna Mendoza

TOTAL POINTS

100 / 100

QUESTION 1

1 Question 1 60 / 60

✓ - **0 pts** Correct

- **10 pts** Patm is wrong
- **20 pts** formulas are wrong
- **5 pts** unit is wrong
- **5 pts** mistake with the formula
- **60 pts** not done
- **10 pts** calculation error

QUESTION 2

2 Question 2 40 / 40

✓ - **0 pts** Correct

- **5 pts** units are missing
- **5 pts** system boundary not labeled
- **40 pts** not done
- **5 pts** P and V values not marked
- **10 pts** state 2 is wrong

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QUIZ 1: 2 problems (10 pts)

Closed book/Closed notes: Show all your work for full credit

1. (6 pts) The average atmospheric pressure on Earth is approximated as a function of altitude by the relation $P_{\text{atm}} = 101.325(1 - 0.02256z)^{5.256}$, where P_{atm} is atmospheric pressure in kPa and z is altitude in km with $z=0$ at sea level. A monometer is used to measure the pressure of a gas in a tank in Denver ($z=1610\text{m}$). The fluid has a density of 850kg/m^3 and the monometer column height is 200cm . Determine the absolute and gage pressure of gas within the tank.
2. (4 pts) A gas contained within a piston-cylinder assembly undergoes two processes in series:

Process 1-2: Constant-pressure compression at 1 bar from $V_1 = 2.5\text{ m}^3$ to $V_2 = 1\text{ m}^3$

Process 2-3: Constant volume compression from 1 bar to 2 bar

- Draw a schematic of the system and label the system boundary.
- Sketch the processes on a p-V diagram; label States 1, 2 and 3 and the pressure and volume values at each numbered state.

① $P_{\text{gage}} = P_{\text{absolute}} - P_{\text{atm}}$

$P_{\text{absolute}} = \rho gh$

$P_{\text{atm}} = 101.325(1 - 0.02256z)^{5.256} \text{ kPa}$

$\rho = 850 \text{ kg/m}^3$

$h = 2 \text{ m}$

$z = 1.61 \text{ km}$

$P_{\text{atm}} = 101.325(1 - 0.02256(1.61))^{5.256}$

$P_{\text{atm}} = 83.42 \text{ kPa} \Rightarrow 83418.79 \text{ Pa}$

$P_{\text{absolute}} = (850 \text{ kg/m}^3)(2 \text{ m})(9.81 \text{ m/s}^2) + 83418.79 \text{ Pa} = 100,095.79 \text{ Pa}$

$P_{\text{gage}} = P_{\text{absolute}} - P_{\text{atm}} = 100,095.79 - 83418.79 = 16677 \text{ Pa}$

$P_{\text{gage}} = 16.68 \text{ kPa}$

②

